TRNG Based on Multiple Entropy Sources Using CTDSM

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Outline

- Introduction
- Literature survey
- Motivation
- STF & NTF of CTDSM
- Non-Idealities and noise analysis
- Proposed design
- Conclusion and future work

Introduction

True Random Number Generator

- Generated through the random physical process.
- Each outcomes carries maximum entropy.
- Non-reproducible.
- Imitates outcome from sequence of a fair coin tosses.

Pseudo Random Number Generator

- Generation based on seeds and algorithms.
- Deterministic, reproducible.
- Sometimes, periodic in long run.

Entropy Sources

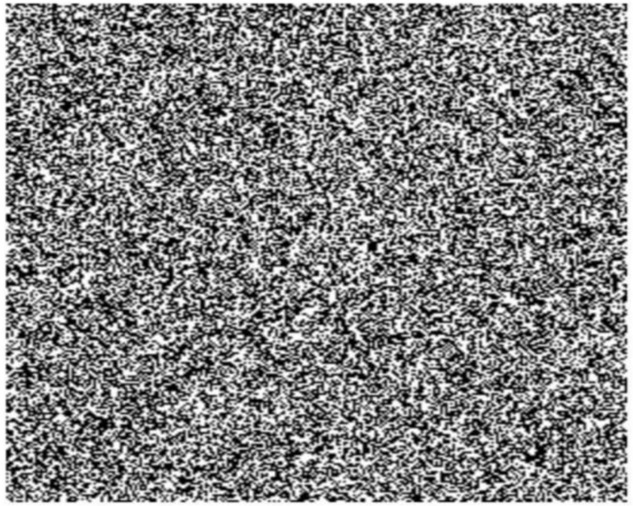
- Device noise, Clock jitter, Metastability, and Chaos.

Applications

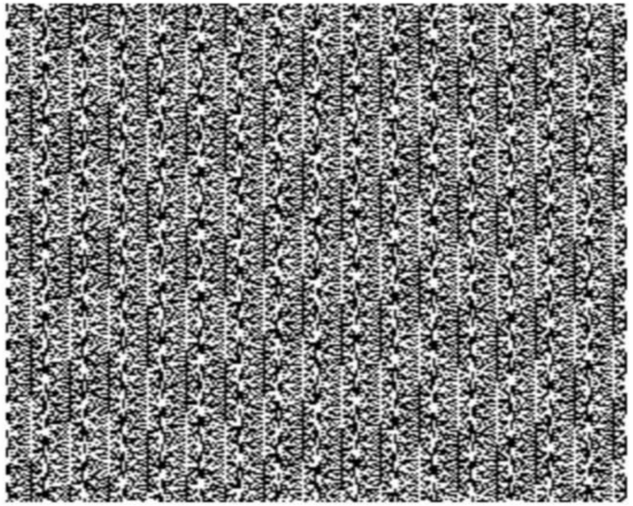
- Critical for Hardware security, Cryptography, Stochastic simulations.

Testing and validation

- NIST SP 800-22 Statistical Test Suit.



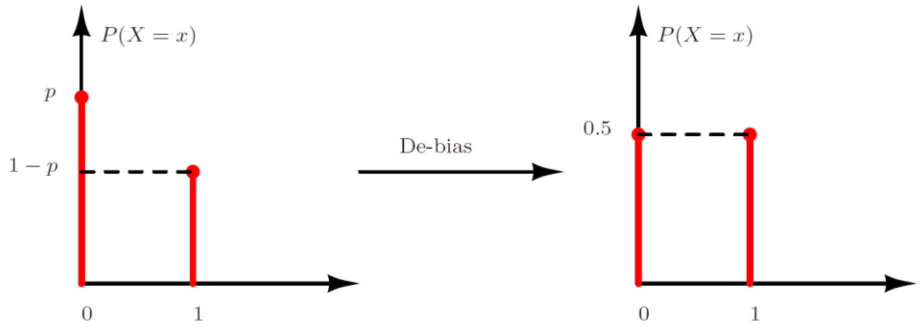
RANDOM.ORG

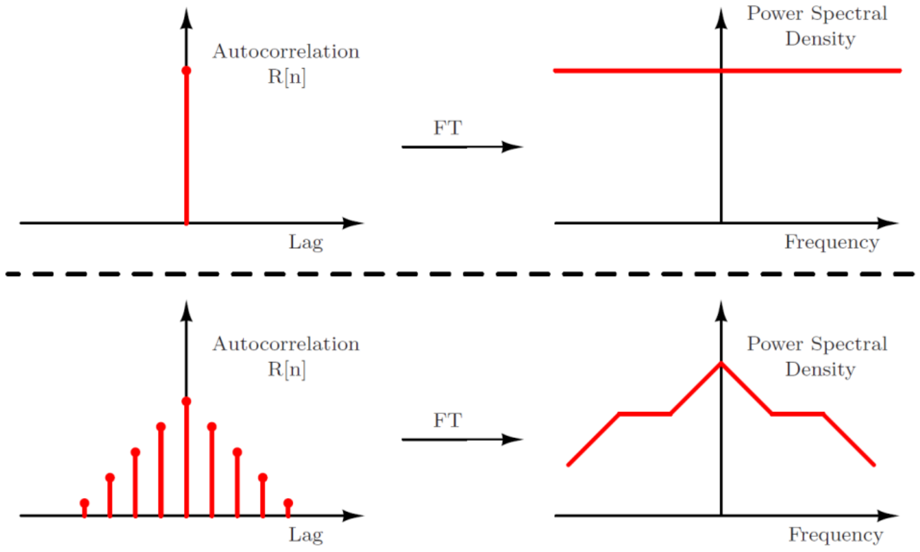


PHP rand() on Microsoft Windows

Bo Allen. Pseudo random vs. true-random. https://boallen.com/random-numbers.html/, February 2012.

Bias and Correlation





Exclusive-OR corrector:

- Leverages piling-up approximation

$$\epsilon\left(X_1 \oplus X_2 \oplus \cdots \oplus X_n\right) = 2^{n-1} \prod_{i=1}^{n} \epsilon\left(X_i\right)$$

Von Neumann corrector:

- 00 and 11: No output,
- 01 and 10: Generates 0 and 1 respectively.
- Perfectly unbiased output.
- Hits Throughput.

Compression based on error-correcting codes.

White random process:

- Unity ACF at zero lag and zero elsewhere.

Colored random process:

- Non-zero ACF at non-zero lags.

Z. Kukorelly. “The piling-up approximation in linear cryptanalysis”. In: IEEE Transactions on Information Theory 47.7 (2001), pp. 2812–2823.

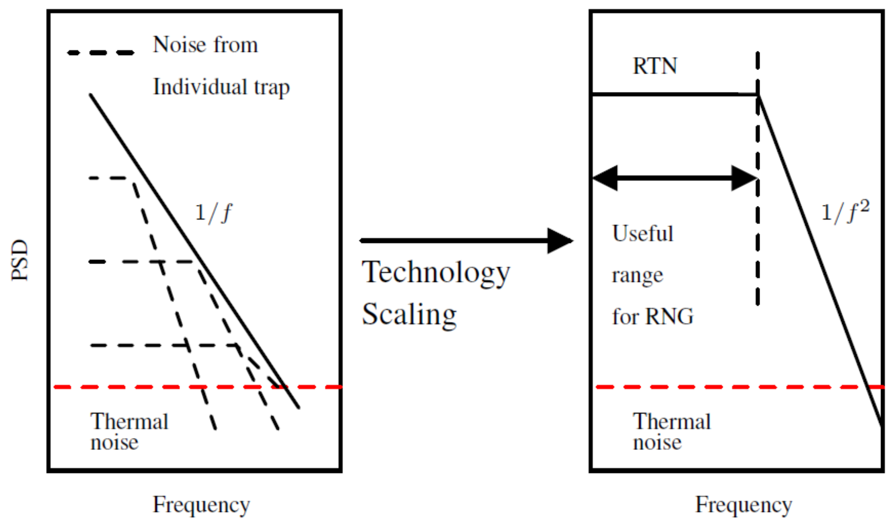
Randomness testing

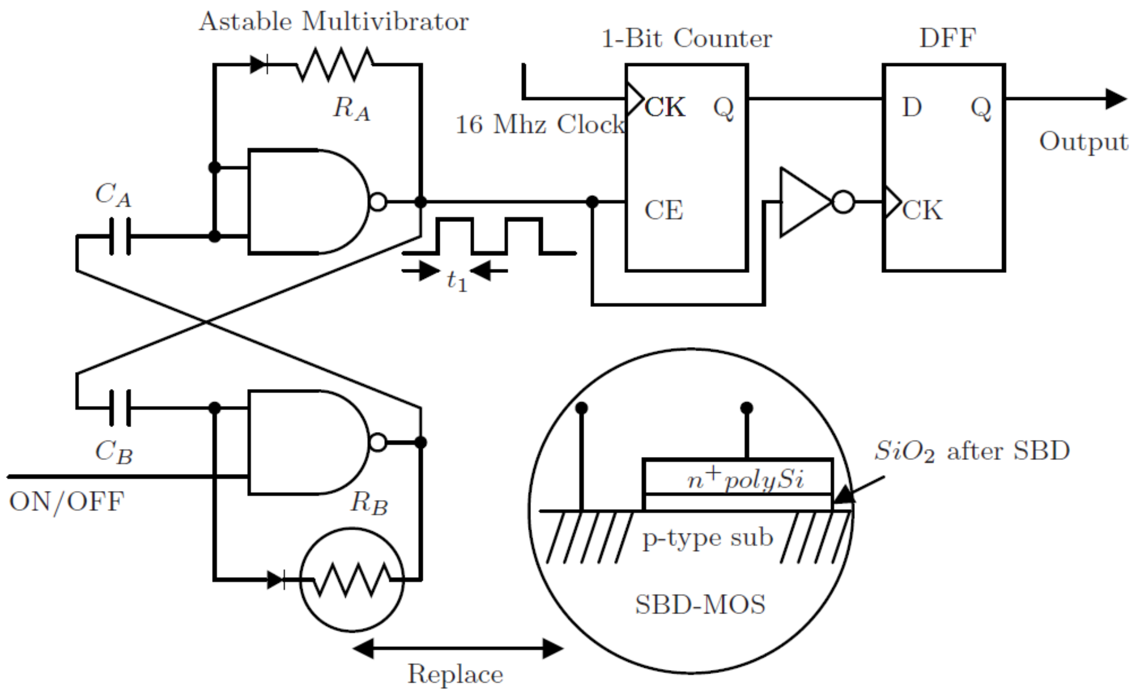
- Autocorrelation and Shannon entropy are commonly used metrics for assessing the quality of randomness.
- Autocorrelation primarily detects linear dependencies but overlooks higher-order relationships or non-linear patterns.
- Shannon entropy measures overall uncertainty but misses localized patterns and assumes independence between symbols.
- The NIST SP 800-22 test suite examines a truly random source’s output for characteristics desirable for use in cryptography.

Test Name	Characteristics
Frequency (Monobit) Test	Checks balance between ones and zeroes
Block Frequency Test	Examines frequency of ones in non-overlapping blocks
Runs Test	Identifies consecutive identical bits (runs)
Longest Run of Ones in a Block Test	Finds the longest run of consecutive ones within a block
Binary Matrix Rank Test	Evaluates sub-matrices rank to detect linear dependencies
Discrete Fourier Transform (Spectral) Test	Analyzes spectral content of the sequence to detect patterns
Non-Overlapping Template Matching Test	Searches for predefined patterns to detect excessive occurrences
Overlapping Template Matching Test	Searches for predefined patterns allowing for overlapping matches
Maurer's Universal Statistical Test	Measures compressibility of the sequence
Linear Complexity Test	Determines randomness by evaluating the length of a LFSR
Serial Test	Examines frequency of overlapping pairs of bit-patterns across the sequence
Approximate Entropy Test	Measures unpredictability by assessing likelihood of repeated patterns
Cumulative Sums (Cusum) Test	Evaluates maximal excursion of the random walk defined by cumulative sum of adjusted digits
Random Excursions Test	Focuses on cycles with specific frequencies in the cumulative sum of the sequence
Random Excursions Variant Test	Examines total number of times a particular state is visited in a cumulative sum random walk

Lawrence E. Bassham et al. “ Sp 800-22 Revision 1a, A Statistical Test Suite for Random and Pseudorandom Number Generators for Cryptographic Applications,” National Institute of Standards and Technology, 2010.

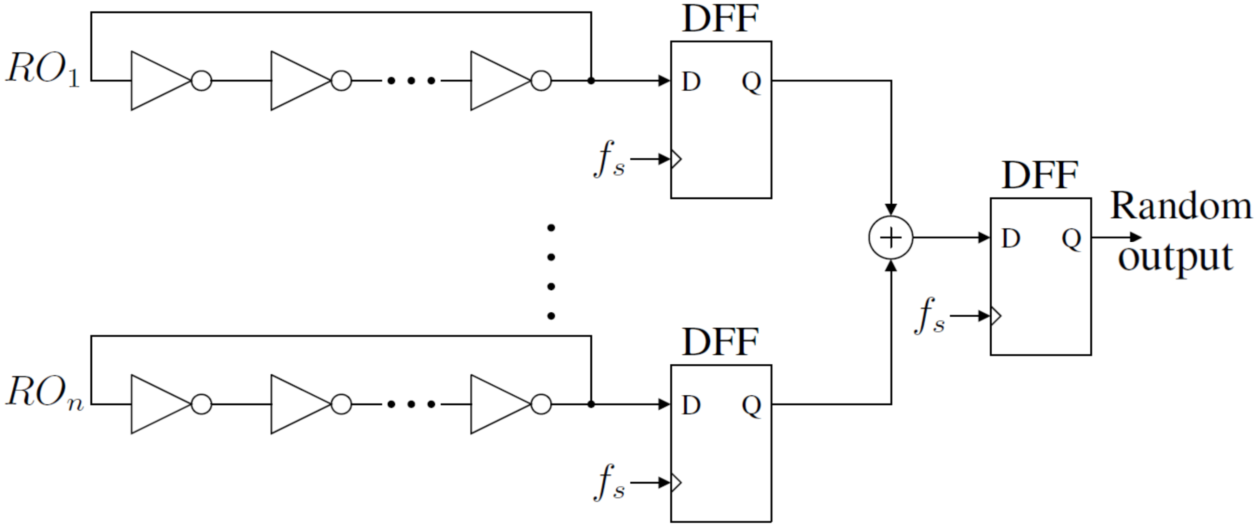
TRNG Based on device noise of MOS

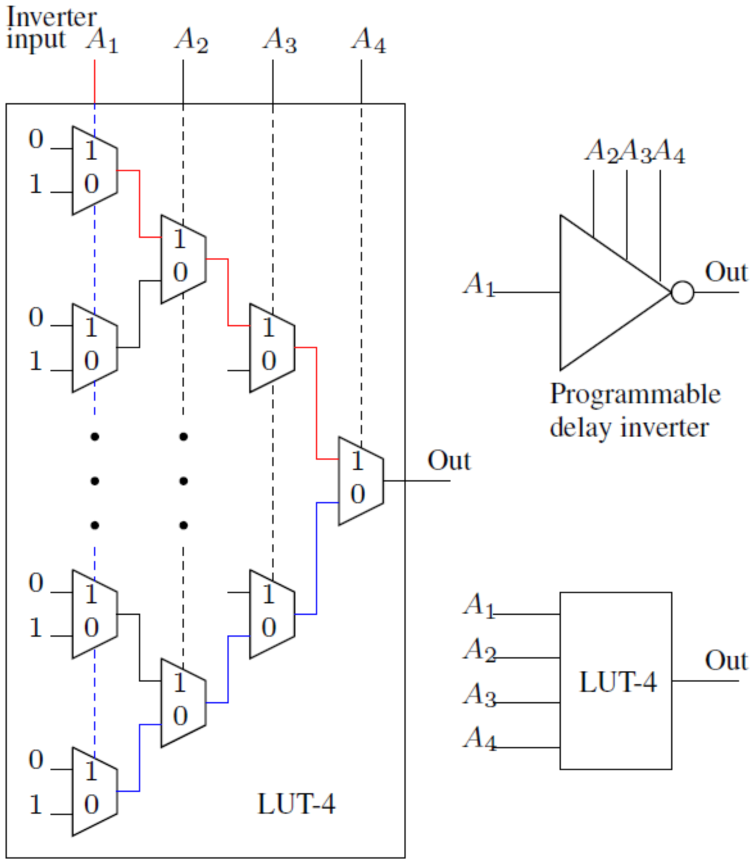




Shinichi Yasuda et al. "Physical random number generator based on MOS structure after soft breakdown". In: IEEE journal of solid-state circuits 39.8 (2004), pp. 1375–1377.

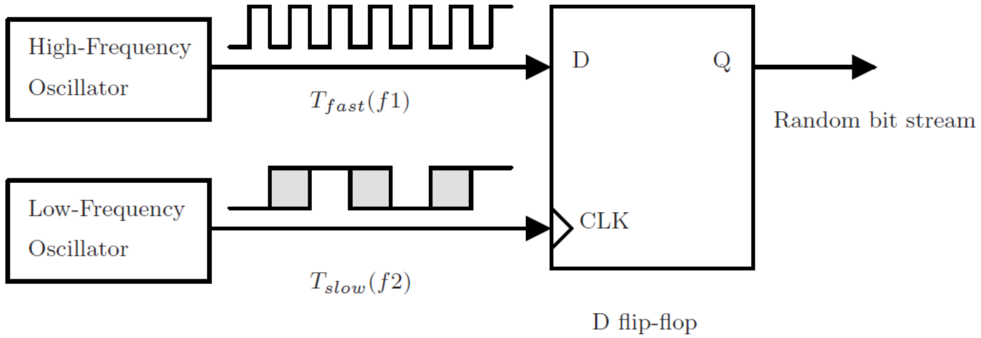
TRNG Based on Coupled Ring Oscillators

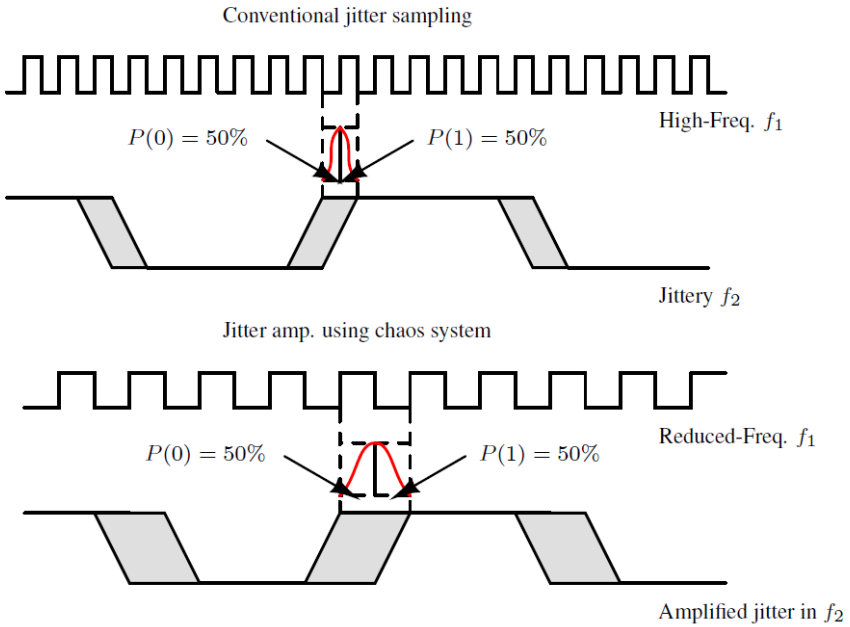


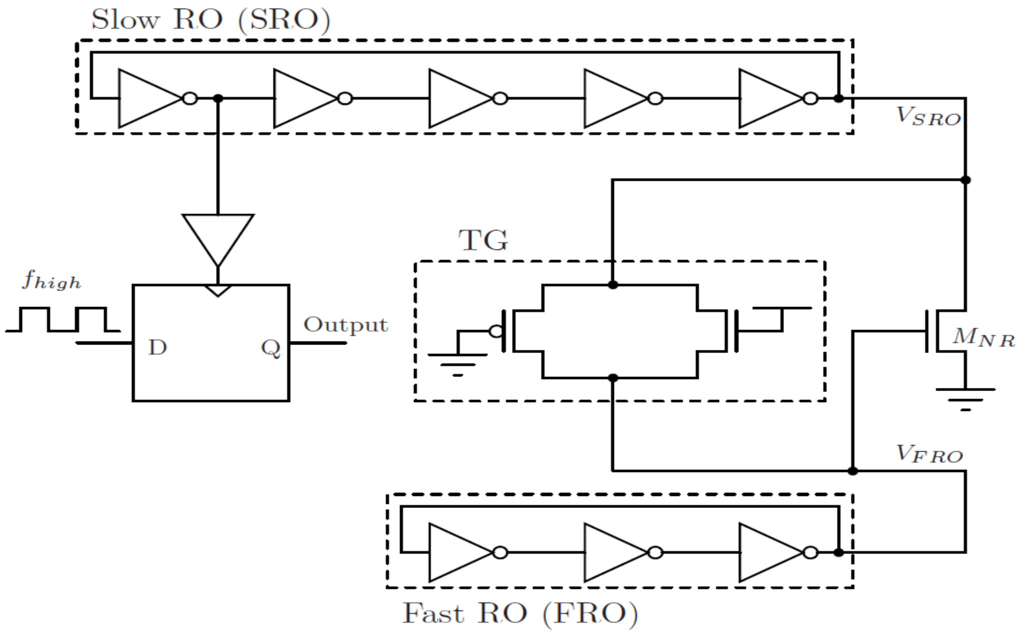


N. Nalla Anandakumar et al. “ FPGA-based true random number generation using programmable delays in oscillator-rings.” In: IEEE Transactions on Circuits and Systems II: Express Briefs, vol. 67, no. 3, pp. 570–574, 2020.

TRNG Based on Chaotic Ring Oscillators

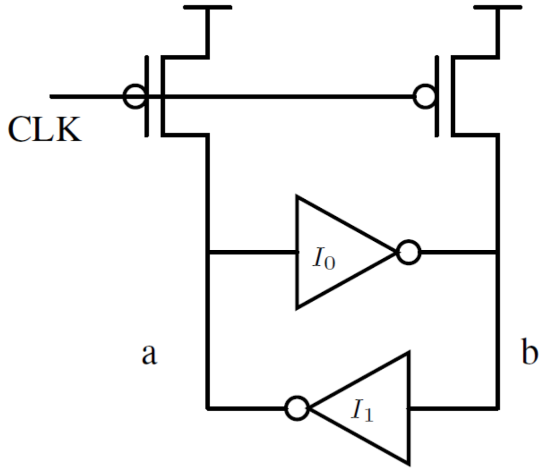


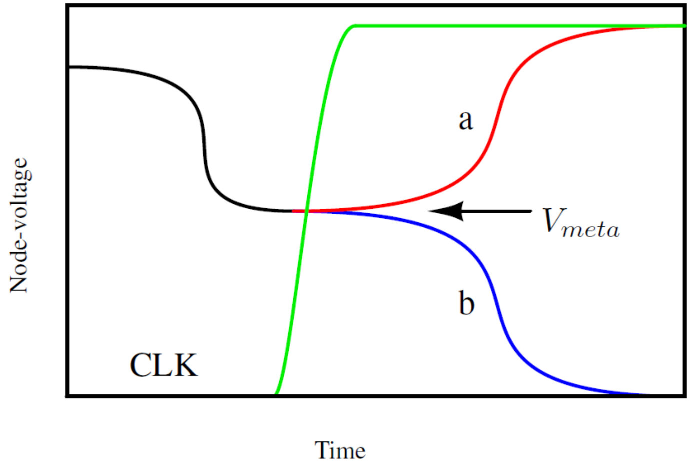


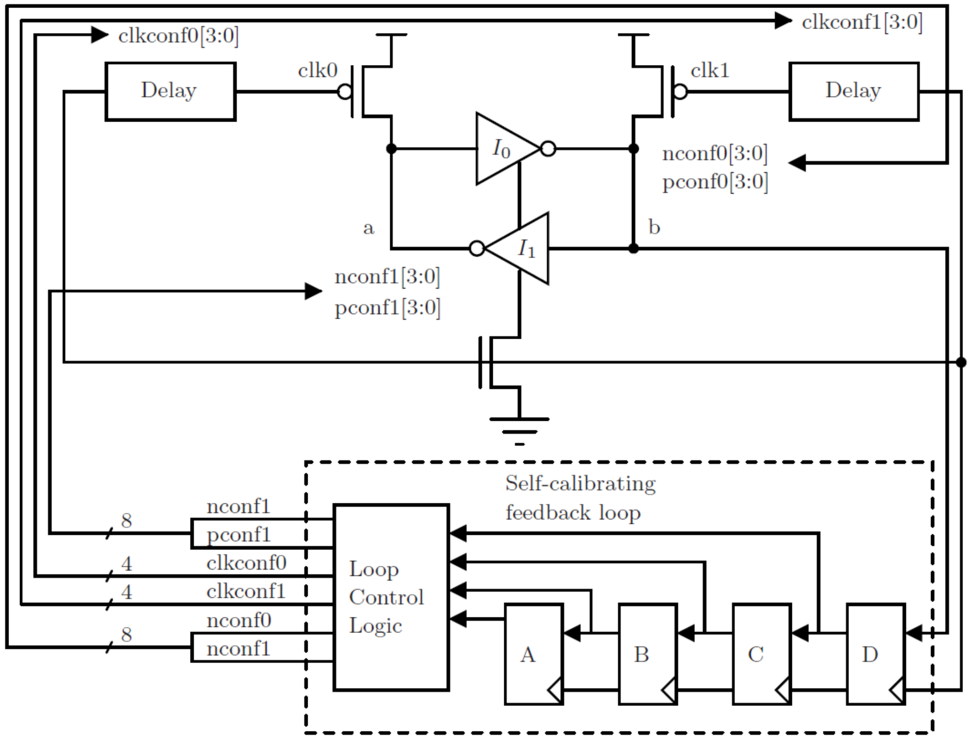


Siva Nishok Dhanuskodi et al. "A chaotic ring oscillator-based random number generator". In: 2014 IEEE International Symposium on Hardware-Oriented Security and Trust (HOST). IEEE. 2014, pp. 160–165.

TRNG Based on Metastability

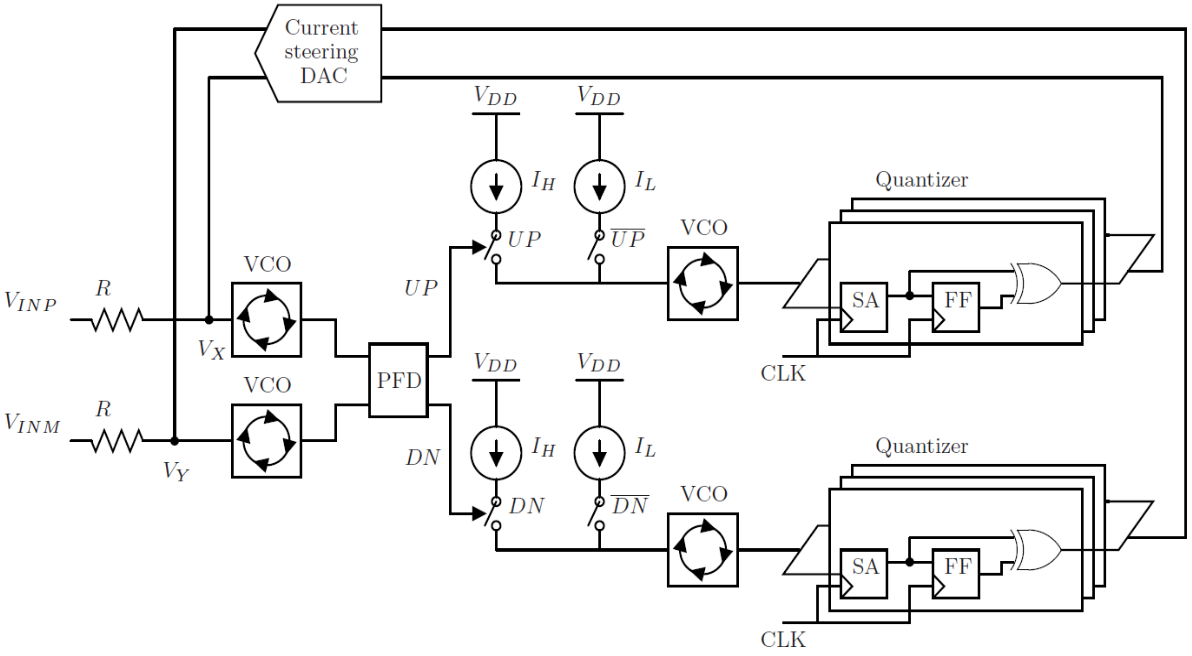


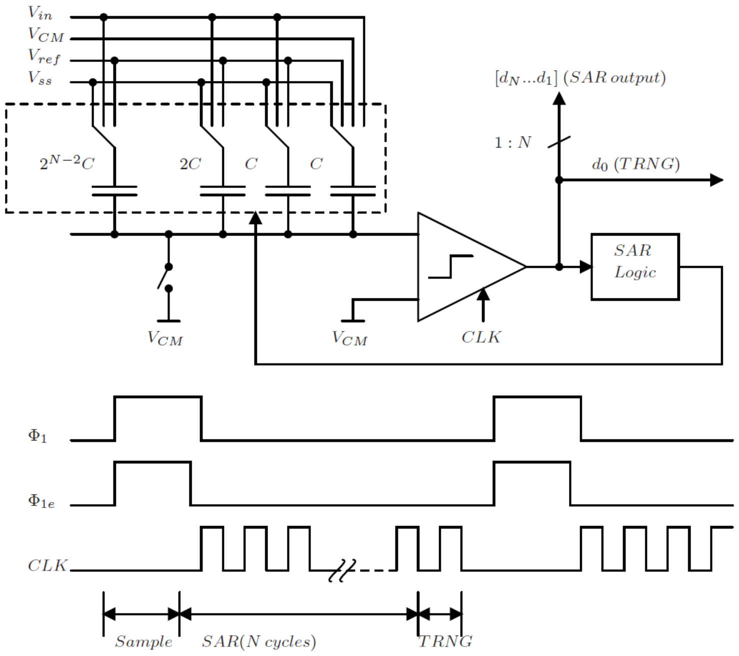




Sanu K Mathew et al. "2.4 Gbps, 7 mW all-digital PVT-variation tolerant true random number generator for 45 nm CMOS high-performance microprocessors". In: IEEE Journal of Solid-State Circuits 47.11 (2012), pp. 2807–2821

TRNG Based on ADC

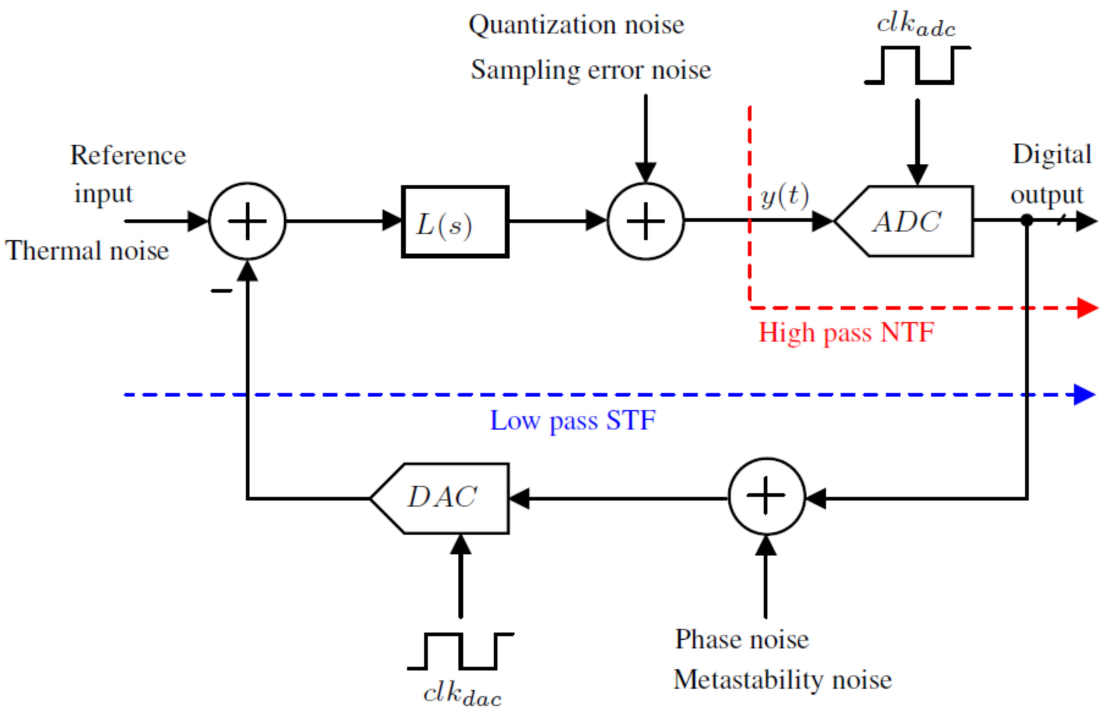




Sanjeev Tannirkulam Chandrasekaran et al. “0.36-mW, 52-Mbps True Random Number Generator Based on a Stochastic Delta–Sigma Modulator”. In: IEEE Solid-State Circuits Letters 3 (2020), pp. 190–193.
Akshay Jayaraj et al. “0.6–1.2 V, 0.22 pJ/bit True Random Number Generator Based on SAR ADC”. In: IEEE Transactions on Circuits and Systems II: Express Briefs 67.10 (2020), pp. 1765–1769.

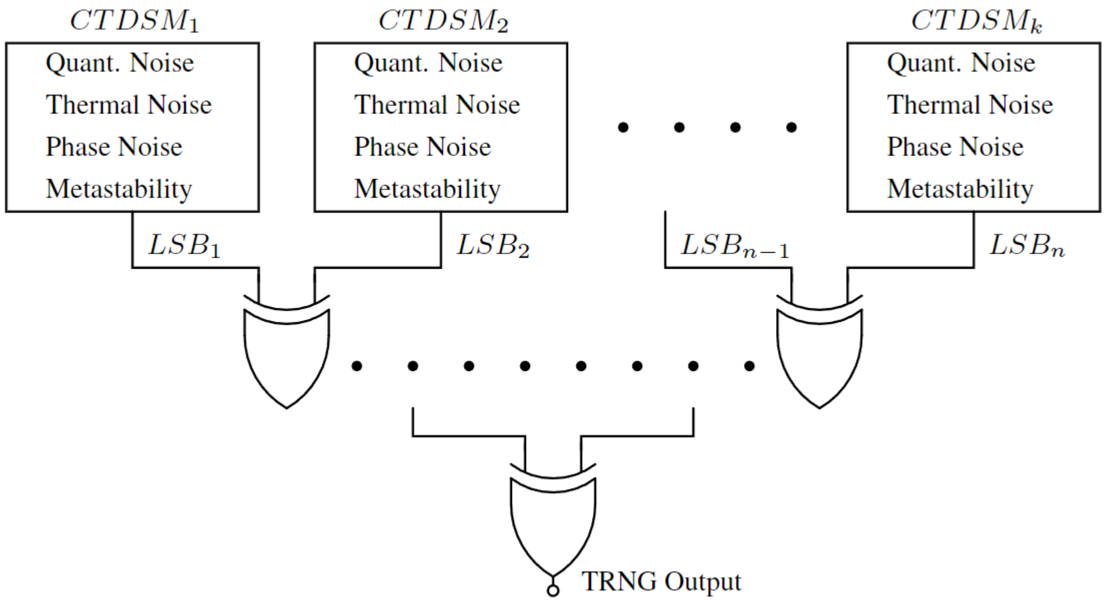
Motivation

- Utilizing a CTDSM as an entropy extractor for TRNG design.
- CTDSM responds to multiple uncorrelated noise such as thermal noise, clock jitter, quantization noise, metastability-induced noise.
- Possibility of intermixing various multiple uncorrelated noise using a single loop.
- Noise from different distribution, characterized by different distributions, converges to a Gaussian distribution, as per the Central Limit Theorem.



TRNG setup

- The proposed TRNG setup utilizes CTDSM to combine entropy from multiple noise source.
- Best way to extract binary random data from entropy-rich CTDSM is by sampling the LSB.
- LSB captures the smallest change, and it is least affected by the offset.
- To generate an unbiased random bit stream, independent and identically distributed samples of LSB from multiple instances of CTDSM are XORed together.



STF & NTF of CTDSM

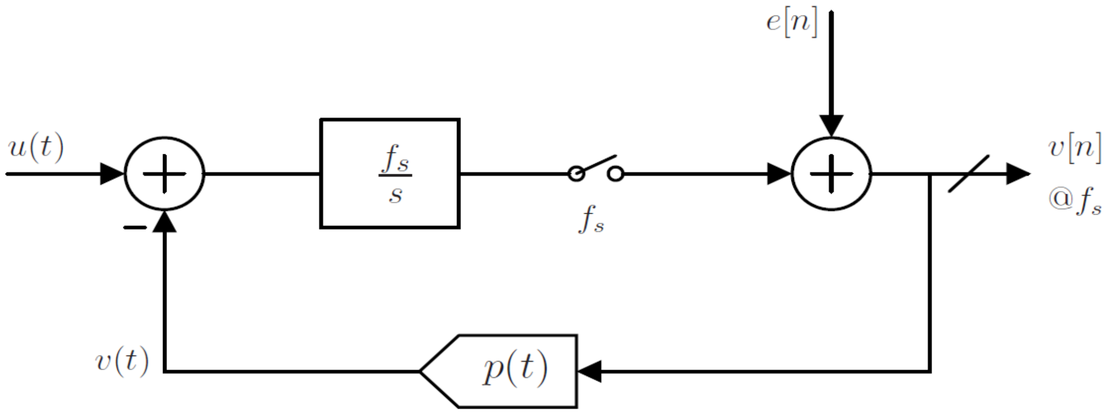


Figure: Block diagram model for CTDSM.

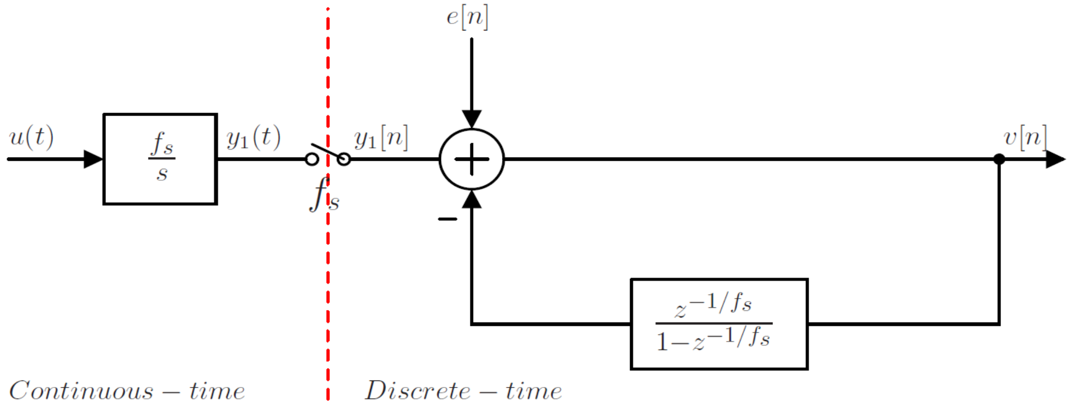


Figure: CTDSM simplified into CT and DT to visualize its STF and NTF.

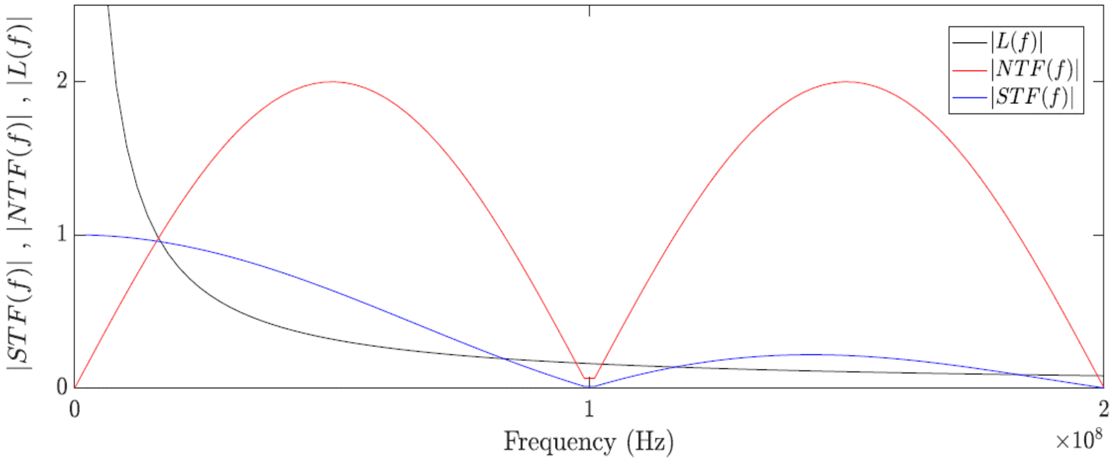


Figure: Magnitude response of Loop-filter, NTF, and the resulting STF of CTDSM.

$$NTF(z) = \frac{V(z)}{E(z)} = \frac{1}{1 - \frac{z^{\frac{-1}{f_s}}}{1 - z^{\frac{-1}{f_s}}}} = \left(1 - z^{\frac{-1}{f_s}}\right)$$

$$\mathrm{STF}(f) = \underbrace{\frac{f_s}{j2\pi f}}_{\text{loop-filter}} \underbrace{\left(1 - e^{-j2\pi\frac{f}{f_s}}\right)}_{NTF} = e^{-j\pi\frac{f}{f_s}} \sin c\left(\frac{f}{f_s}\right)$$

Shanthi Pavan Richard Schreie Gabor Temes. Understanding Delta-Sigma Data Converters. Second Edition. IEEE Press, Wiley, 2017.

Noise analysis:

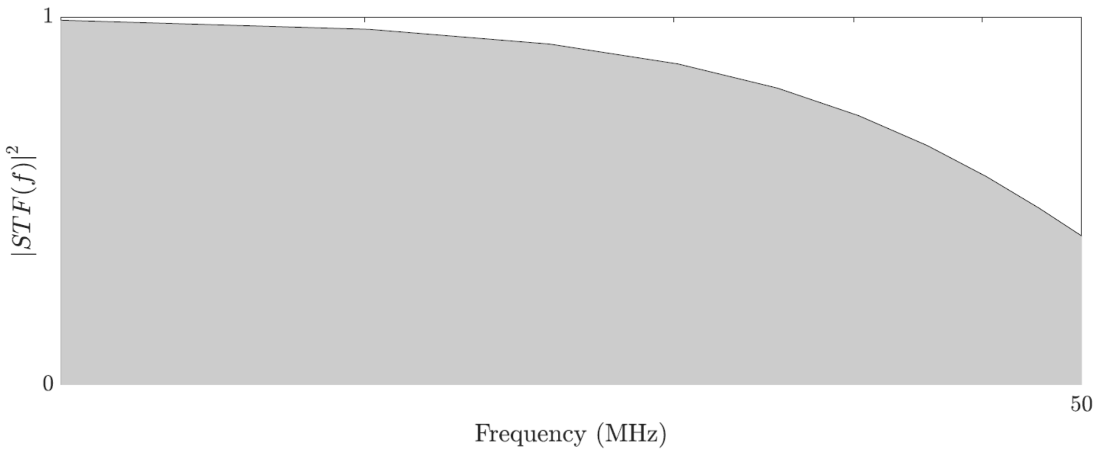


Figure: PSD of thermal noise appearing at the output of the CTDSM.

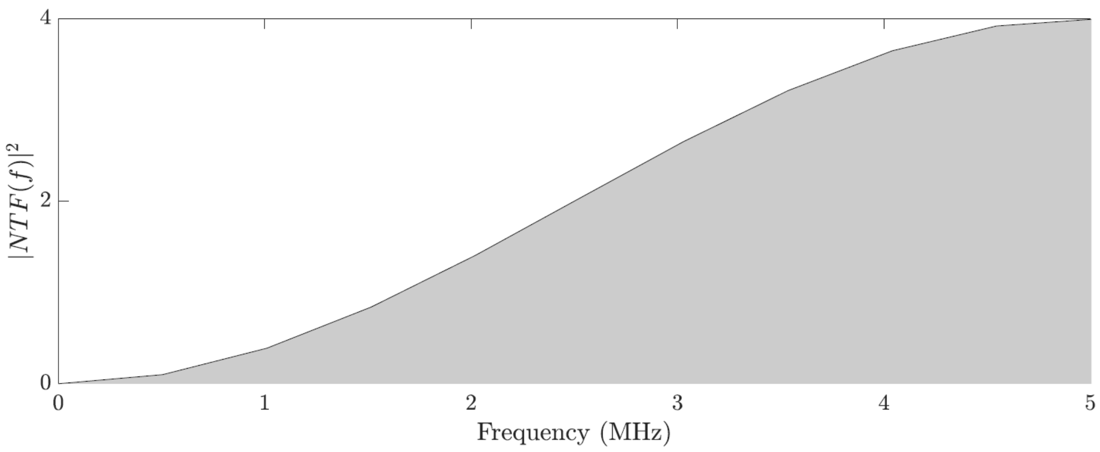


Figure: PSD of high-pass shaped quantization noise appearing at the output of the CTDSM.

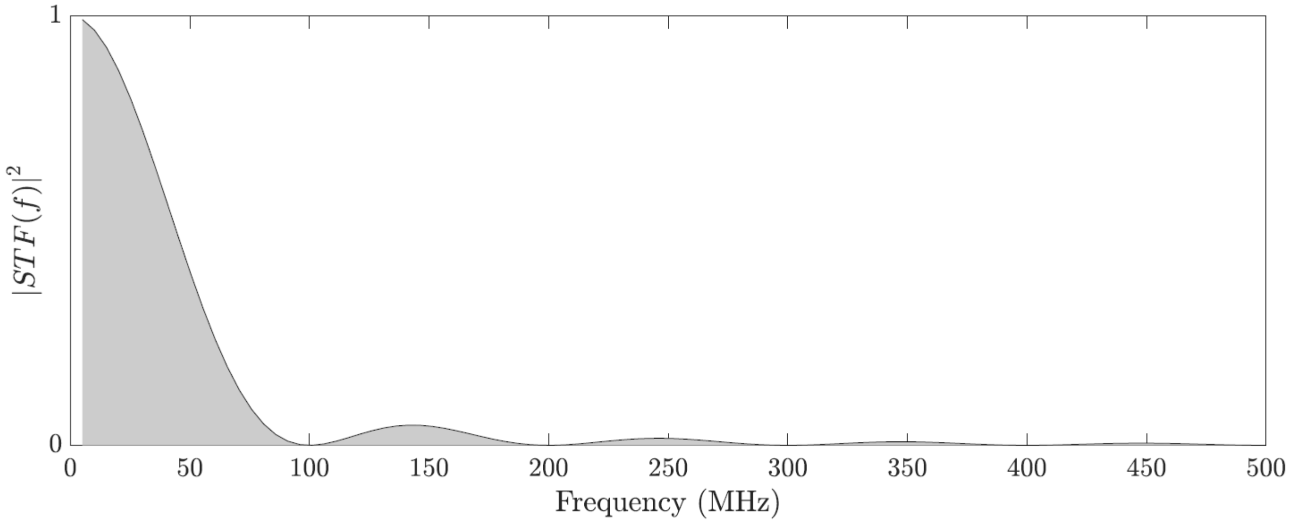
- Passes through low-pass STF, appears directly at the output.
- Potential to alias back.

$$S_{v_n i} = 4kTR \quad V^2/Hz$$
$$\sigma^2_{v_n o} = \int_0^{\frac{f_s}{2}} S_{v_n i} \cdot |(\mathrm{STF}(f))|^2 \, df \quad V^2$$

- Quantization noise is assumed to be additive, white, uniformly distributed and independent of the input for busy input signals.
- The additive quantization noise spectrum is shaped by high-pass NTF.

$$S_{q_n i} = \frac{\Delta^2}{12} \cdot \left(\frac{2}{f_s}\right) \quad V^2/Hz$$
$$\sigma^2_{q_n o} = \int_0^{\frac{f_s}{2}} S_{q_n i} \cdot |(\mathrm{NTF}(f))|^2 \, df \quad V^2$$

Noise analysis: Thermal noise(cont.)



$K \cdot f_s \pm f_s/2$	$\sigma^2_{v_n alias}$ **at** $K \cdot f_s \pm f_s/2$	$\sigma^2_{v_n o}$
$100MHz \pm 50MHz$	$3.14 \times 10^{-4}\ V^2$	$1.85 \times 10^{-3}\ V^2$
$200MHz \pm 50MHz$	$5.60 \times 10^{-5}\ V^2$	$1.90 \times 10^{-3}\ V^2$
$300MHz \pm 50MHz$	$2.35 \times 10^{-5}\ V^2$	$1.92 \times 10^{-3}\ V^2$
$400MHz \pm 50MHz$	$1.29 \times 10^{-5}\ V^2$	$1.93 \times 10^{-3}\ V^2$
$500MHz \pm 50MHz$	$8.20 \times 10^{-6}\ V^2$	$1.93 \times 10^{-3}\ V^2$
$600MHz \pm 50MHz$	$5.68 \times 10^{-6}\ V^2$	$1.94 \times 10^{-3}\ V^2$
$700MHz \pm 50MHz$	.	$1.94 \times 10^{-3}\ V^2$
$800MHz \pm 50MHz$	.	$1.94 \times 10^{-3}\ V^2$
$900MHz \pm 50MHz$	.	$1.94 \times 10^{-3}\ V^2$
$1000MHz \pm 50MHz$	.	$1.94 \times 10^{-3}\ V^2$

- Considering ideal sampling, mean square value of thermal noise that can be aliased back in CTDSM is calculated.
- No significant contribution incase of NRZ DAC due to inherent alias rejection property of CTDSM.

Clock jitter effect on CTDSM

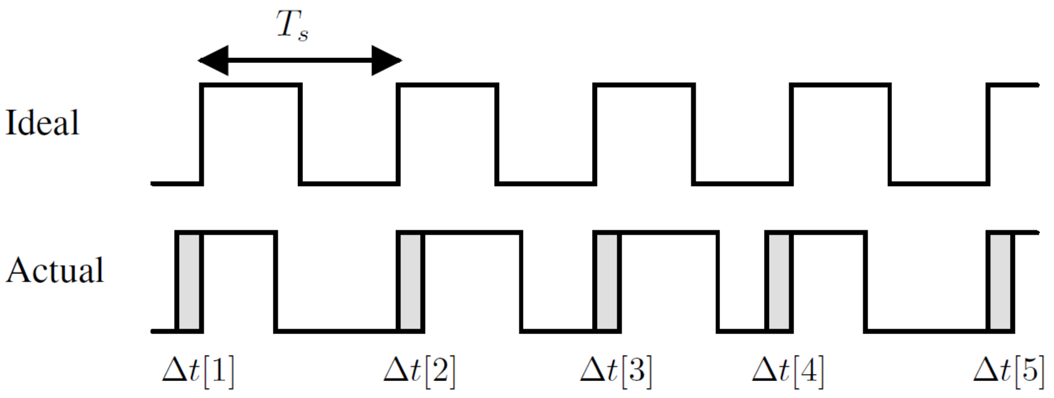


Figure 3.16: White uncorrelated error sequence due to clock jitter.

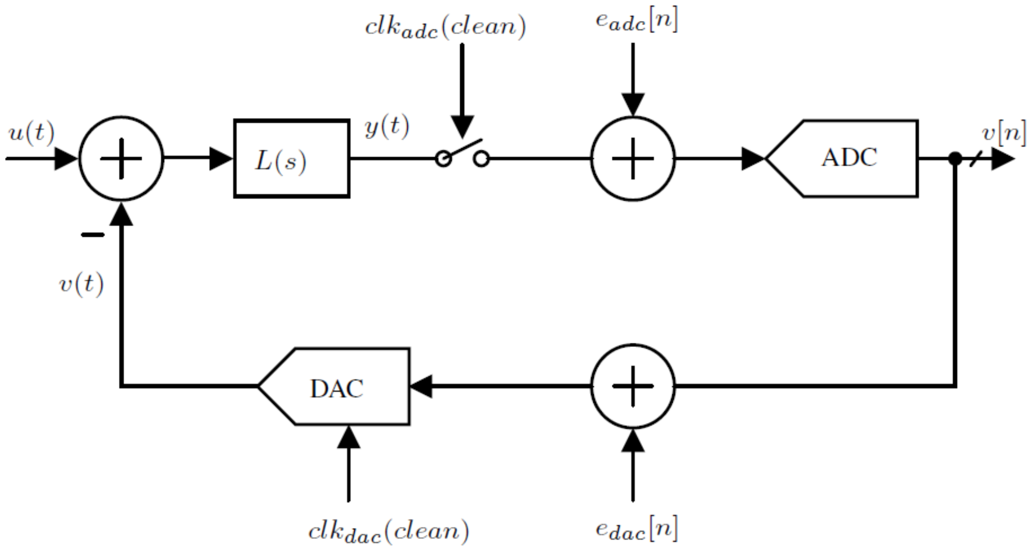


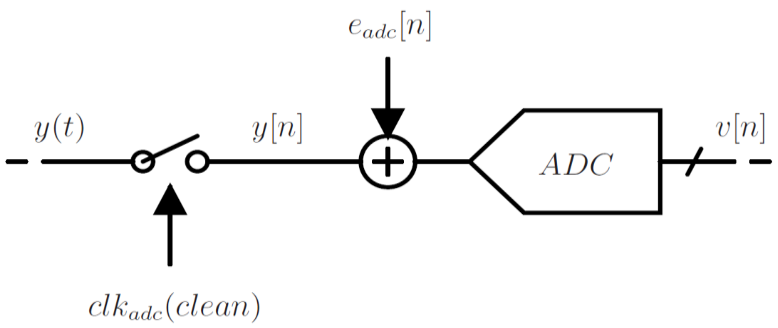
Figure: Jitter induced error in a CTDSM.

- Timing errors sequence Δt[n], assumed to be uncorrelated, Gaussian distributed and white with RMS value of $\sigma_{\Delta t}$.
- clk_{adc} and clk_{dac} introduces error e_{adc}[n] and e_{dac}[n] due to jitter.
- e_{adc}[n] sees NTF and gets high-pass shaped.
- e_{dac}[n] passes through STF and appears as it is at output.

J.A. Cherry et al. “Clock jitter and quantizer metastability in continuous-time delta-sigma modulators”. In: IEEE Transactions on Circuits and Systems II: Analog and Digital Signal Processing 46.6 (1999), pp. 661–676.

Noise analysis: Phase noise of clk_{adc}

1.

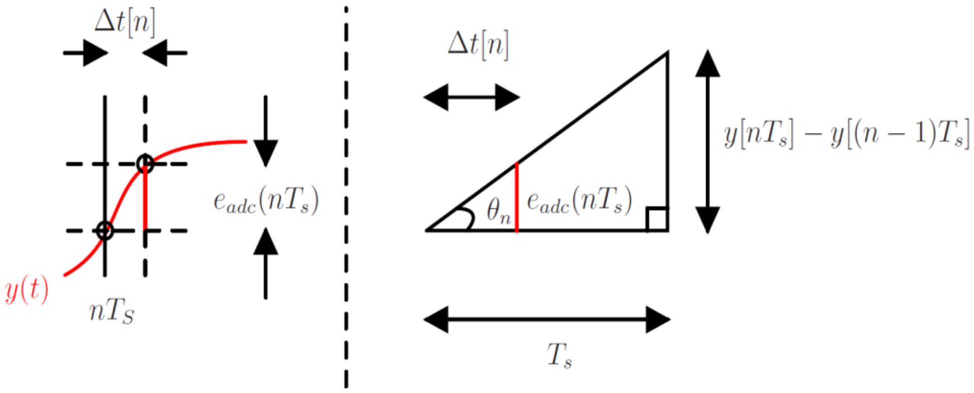


$$y(t) \rightarrow y\left(nT_s\right) \{\text{Ideal } clk_{adc}\}$$

$$y(t) \rightarrow y\left(nT_s + \Delta t[n]\right) \approx y\left(nT_s\right) + \Delta t[n] \cdot \left.\frac{dy}{dt}\right|_{nT_s} \{\text{Jittery } clk_{adc}\}$$

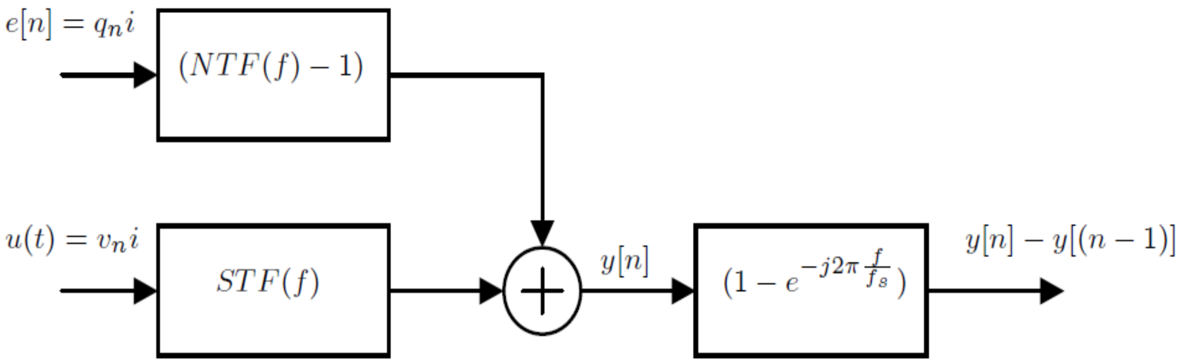
$$e_{adc}[n] = y\left(nT_s + \Delta t[n]\right) - y\left(nT_s\right) = \Delta t[n] \cdot \left.\frac{dy}{dt}\right|_{nT_s}$$

2.



$$e_{adc}[n] = \Delta t[n] \cdot tan\left(\theta_n\right) = \Delta t[n] \cdot \frac{y[n]-y[n-1]}{T_s}$$

3.



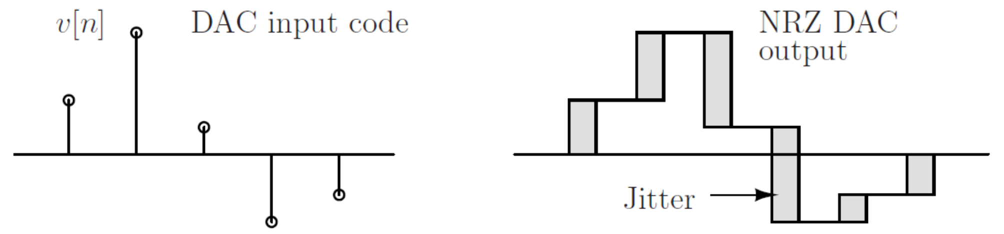
4.

$$S_{e_{adc}i} = \sigma^2_{e_{adc}i} \cdot \left(\frac{2}{f_s}\right) \quad V^2/Hz$$

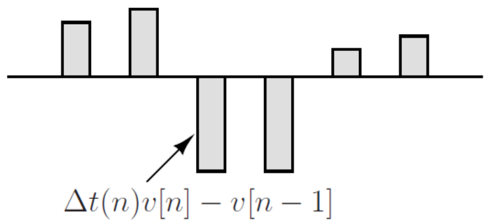
$$\sigma^2_{e_{adc}o} = \int_0^{\frac{f_s}{2}} S_{e_{adc}i} \cdot \left|(\text{NTF}(f))\right|^2 df \quad V^2$$

Noise analysis: Phase noise of clk_{dac}

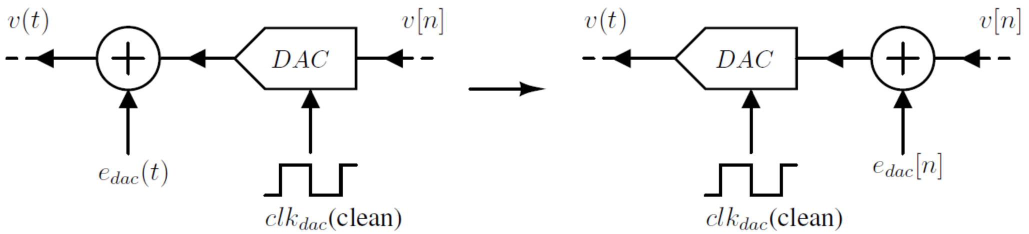
1.



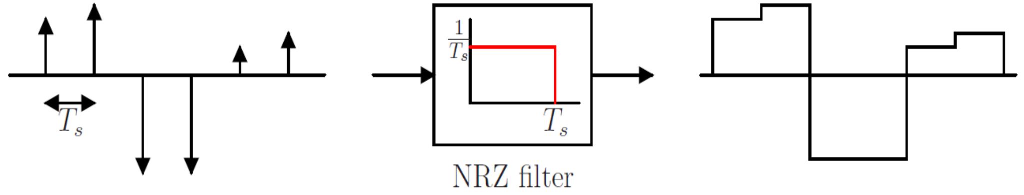
$$e_{dac}(t) = \sum_n \Delta t(n)(v[n] - v[n-1])\delta(t - nT_s)$$



2.

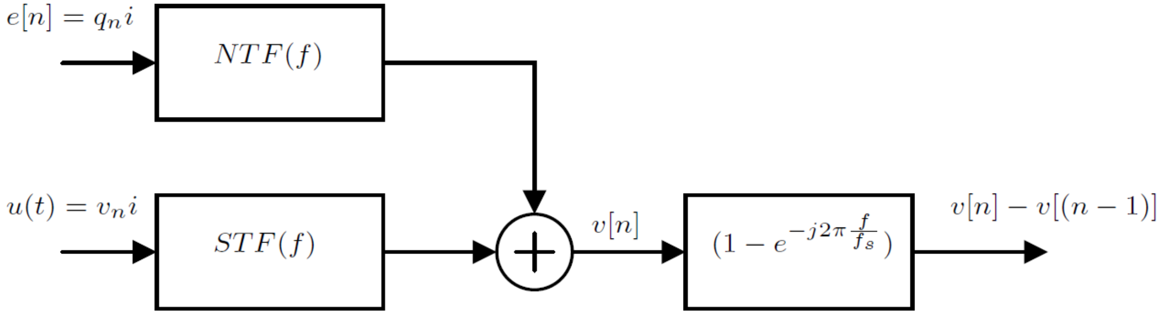


$$e_{dac}(t) = \sum_n \Delta t(n)(v[n] - v[n-1])\delta(t - nT_s) \qquad e_{dac}[n] = \frac{\Delta t(n)(v[n]-v[n-1])}{T_s}$$



$$e_{dac}[n] = \frac{\Delta t[n]}{T_s} \cdot (v[n] - v[n-1])$$

3.

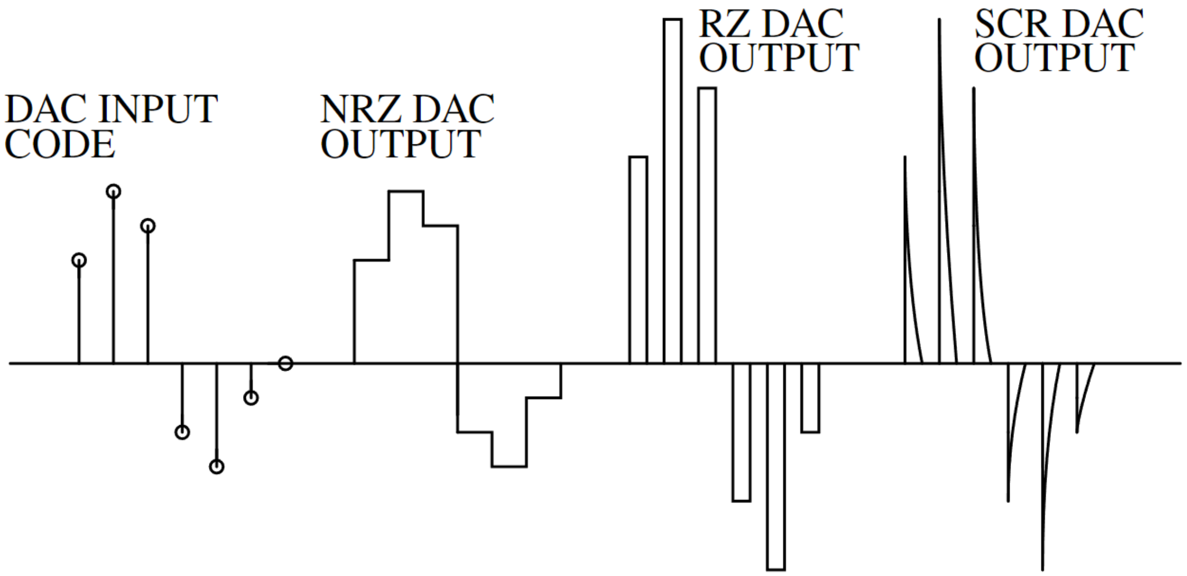


4.

$$\sigma^2_{e_{dac}} = \frac{\sigma^2_{\Delta t}}{T_s^2} \int_0^{\frac{f_s}{2}} \left| S_{v_n i} \cdot STF(f) \cdot \left(1 - e^{-j2\pi \frac{f}{f_s}}\right) \right|^2 df$$
$$+ \frac{\sigma^2_{\Delta t}}{T_s^2} \int_0^{\frac{f_s}{2}} \left| S_{q_n i} \cdot NTF(f) \cdot \left(1 - e^{-j2\pi \frac{f}{f_s}}\right) \right|^2 df \quad V^2$$

Shanthi Pavan Richard Schreie Gabor Temes. Understanding Delta-Sigma Data Converters. Second Edition. IEEE Press, Wiley, 2017.

Noise analysis: Phase noise of clk_{dac} (cont.)



NRZ DAC:

$$e_{dac}[n] = (v[n] - v[n-1])\frac{\Delta t[n]}{T_s}$$

RZ DAC:

$$e_{dac}[n] = (2v[n])\frac{(\Delta t_{rise}[n] + \Delta t_{fall}[n])}{T_s}$$

SCR DAC:

$$e_{dac}[n] = \frac{v[n]}{R_d}\exp\left(\frac{-T_s}{2R_dC_d}\right)\left(\Delta t[n] - \Delta t\left[n+\frac{1}{2}\right]\right)$$

- Poor alias rejection in case of RZ DAC.
- Strong rejection at $(k.f_s + \Delta f)$, where $\Delta f << f_s$ for NRZ DAC.

$$|STF\left(j2\pi\left(kf_s + \Delta f\right)\right)| = \left|\frac{L\left(j2\pi\left(kf_s + \Delta f\right)\right)}{L(j2\pi\Delta f)}\right|$$
$$= \left|\frac{\Delta f}{kf_s + \Delta f}\right|$$

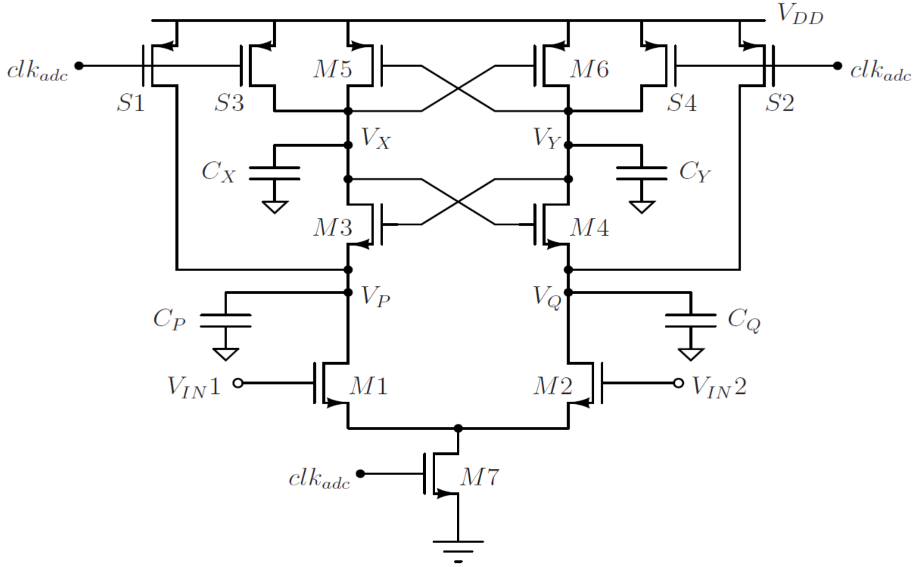
- STF can be seen degraded by practical G_{ota} for RZ DAC.

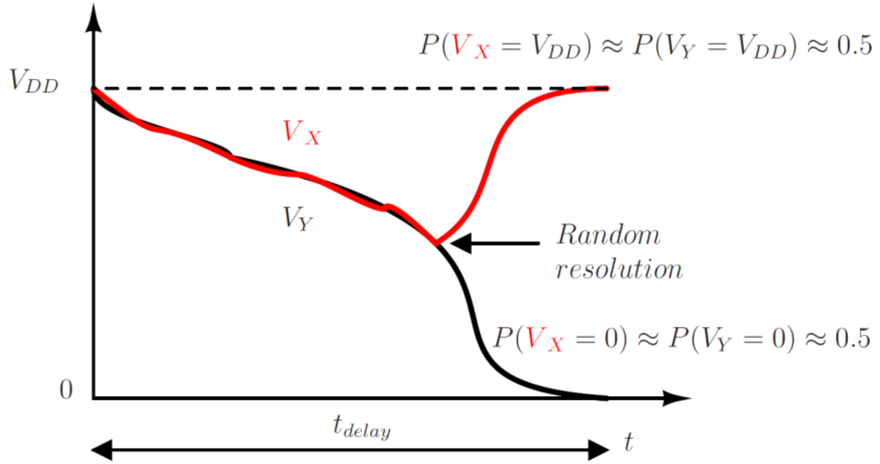
$$|STF(j2\pi(kf_s + \Delta f))| \approx \frac{f_sC}{G_{ota}}$$

Shanthi Pavan. “Alias Rejection of Continuous-Time ΔΣ Modulators With Switched-Capacitor Feedback DACs”. In: IEEE Transactions on Circuits and Systems I: Regular Papers 58.2 (2011), pp. 233–243.

Metastability in comparator

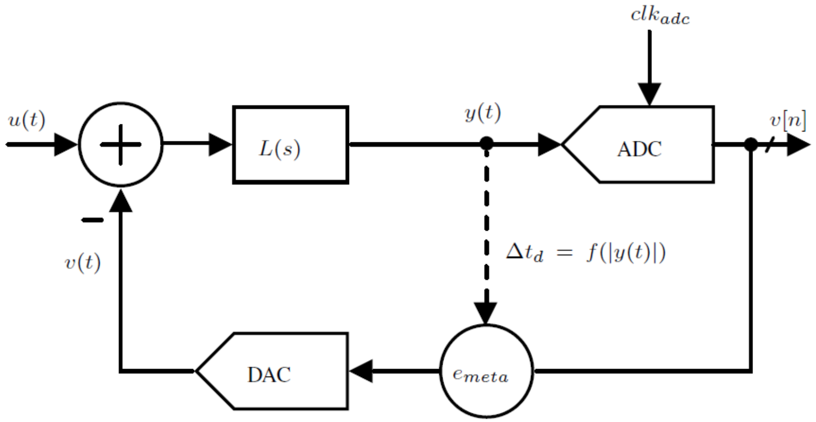
1.





$$t_{\text{delay}} \approx \tau \ln\left(\frac{\alpha V_{DD}}{\beta |V_{IN}1 - V_{IN}2|}\right) \quad where,\ \tau = \frac{C_{X,Y}}{g_{m3,4}\,(1 - C_{X,Y}/C_{P,Q})}$$

2.



3.

$$e_{meta}[n] = \frac{t_{delay}}{T_s}(v[n] - v[n-1])$$

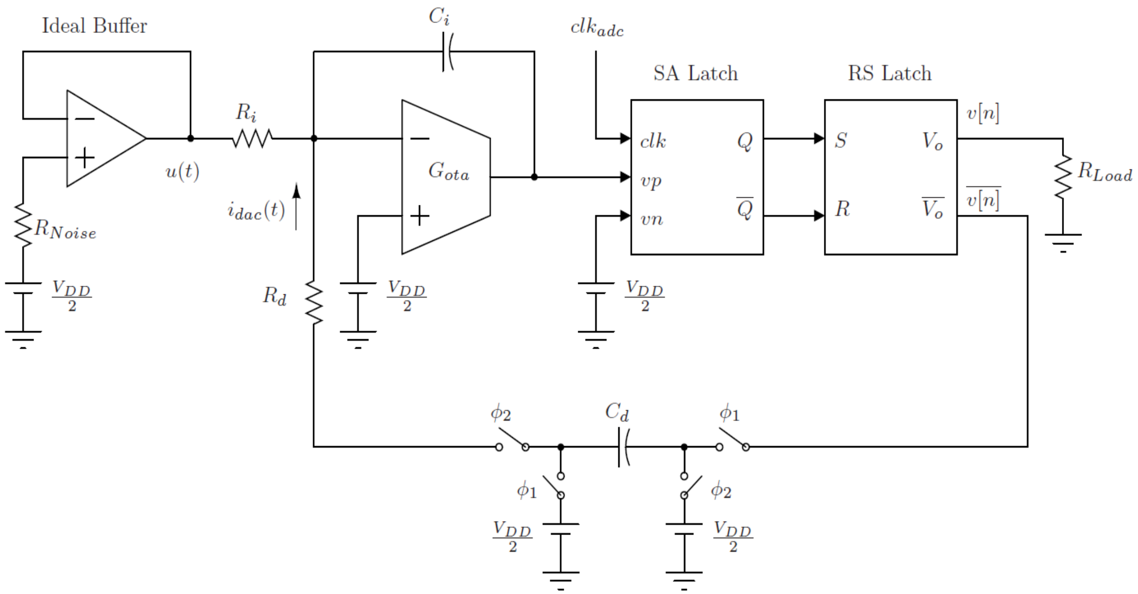
Behzad Razavi. “The StrongARM Latch [A Circuit for All Seasons]”. In: IEEE Solid-State Circuits Magazine 7.2 (2015), pp. 12–17.
Shanthi Pavan Richard Schreie Gabor Temes. Understanding Delta-Sigma Data Converters. Second Edition. IEEE Press, Wiley, 2017.

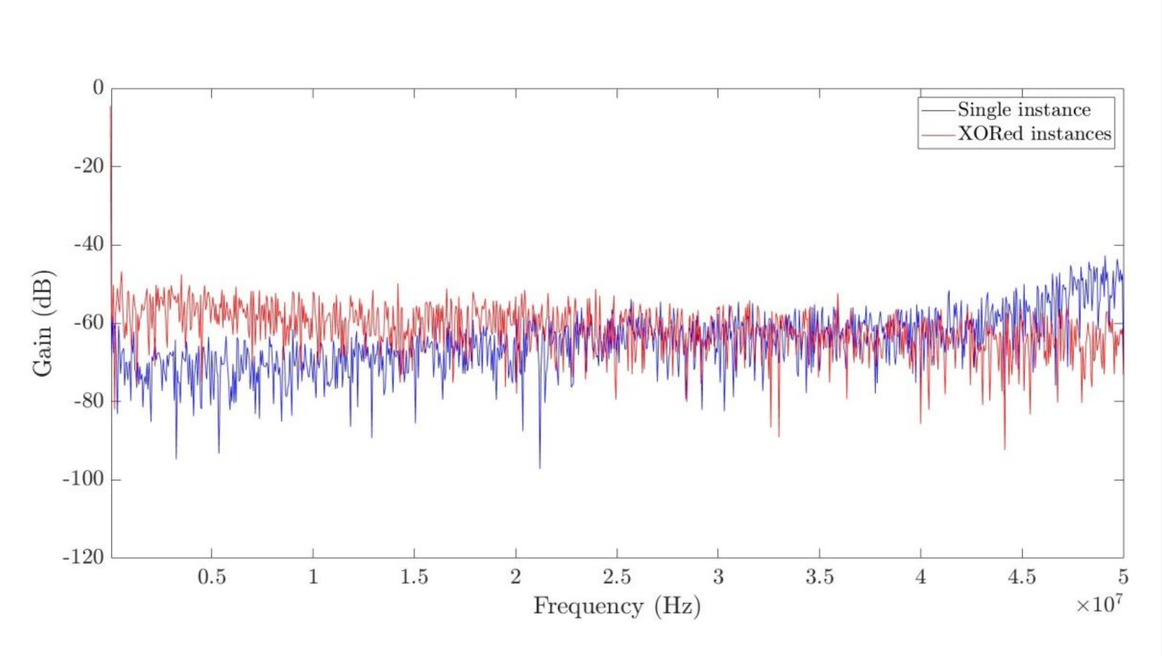
Noise analysis summary

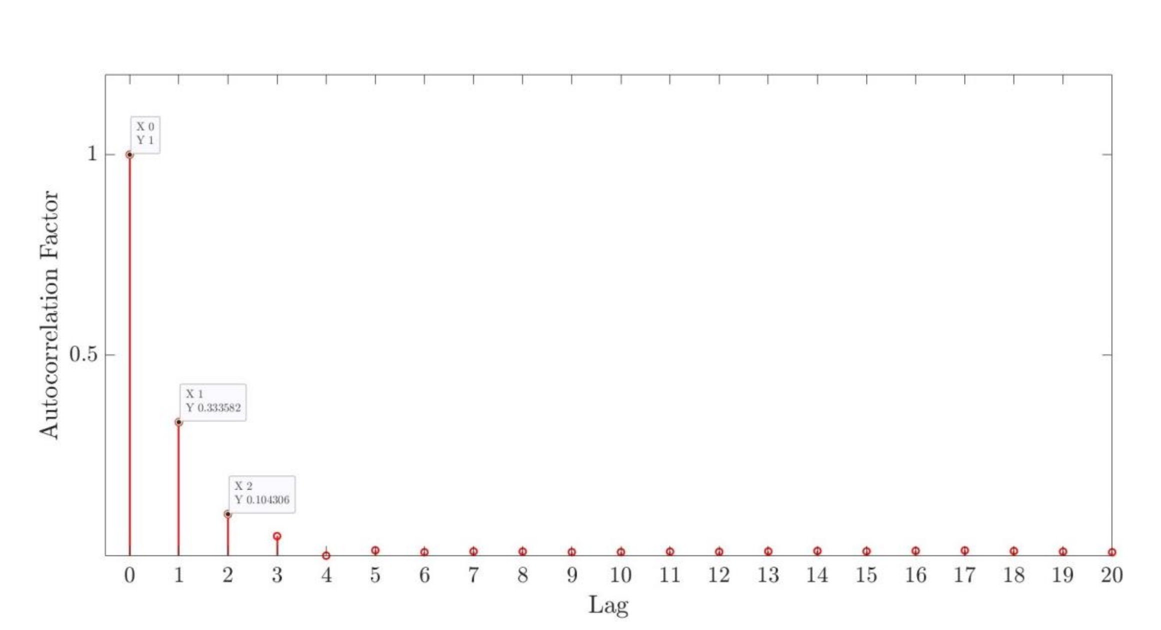
Noise sources	**1st Order 1 Bit**	**1st Order 4 Bit**
Quantization Noise ($q_n i$)	$60\ mV^2$	$937.5\ \mu V^2$
Thermal Noise ($v_n i$)	$1.94\ mV^2$	$1940\ \mu V^2$
Phase Noise ($e_{adc}[n]$)	$5\ mV^2$	$281\ \mu V^2$
Phase Noise ($e_{dac}[n]$)	$9.1\ mV^2$	$269.3\ \mu V^2$
Metastability ($e_{meta}[n]$)	$0.3\ mV^2$	$10.7\ \mu V^2$

- For the 1st-order 1-bit modulator, the overall noise power is dominated by quantization noise.
- The quantization noise has to be reduced significantly.
- By incorporating a 4-bit quantizer in the delta-sigma loop, we observe a more balanced distribution of noise.

1-bit CTDSM based TRNG

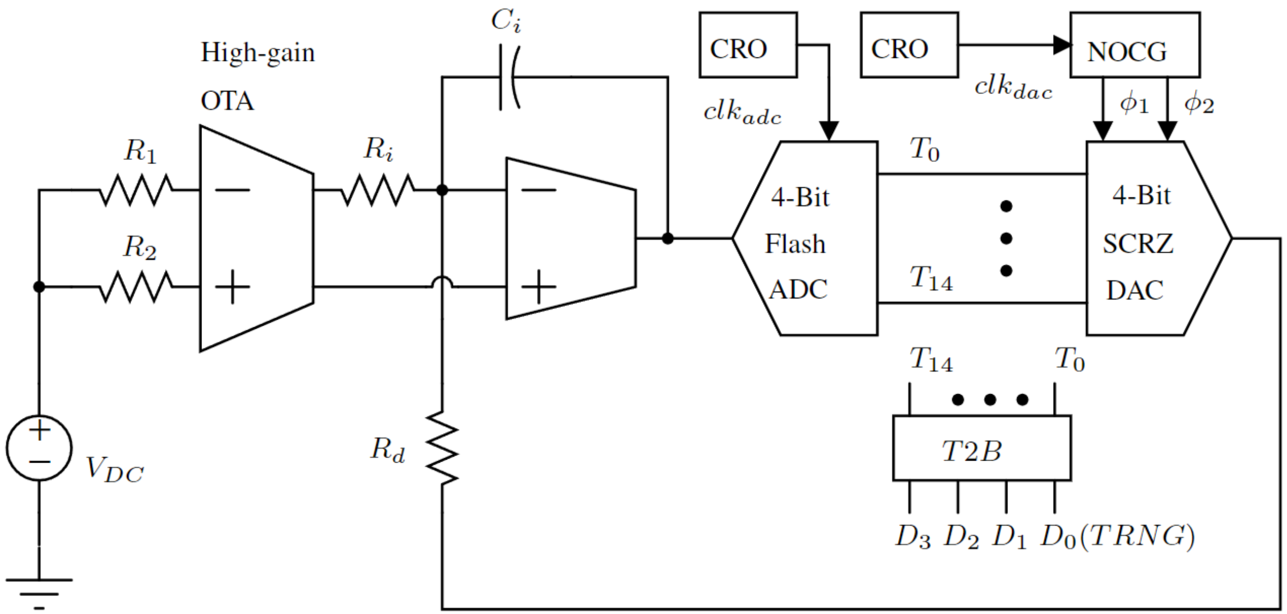






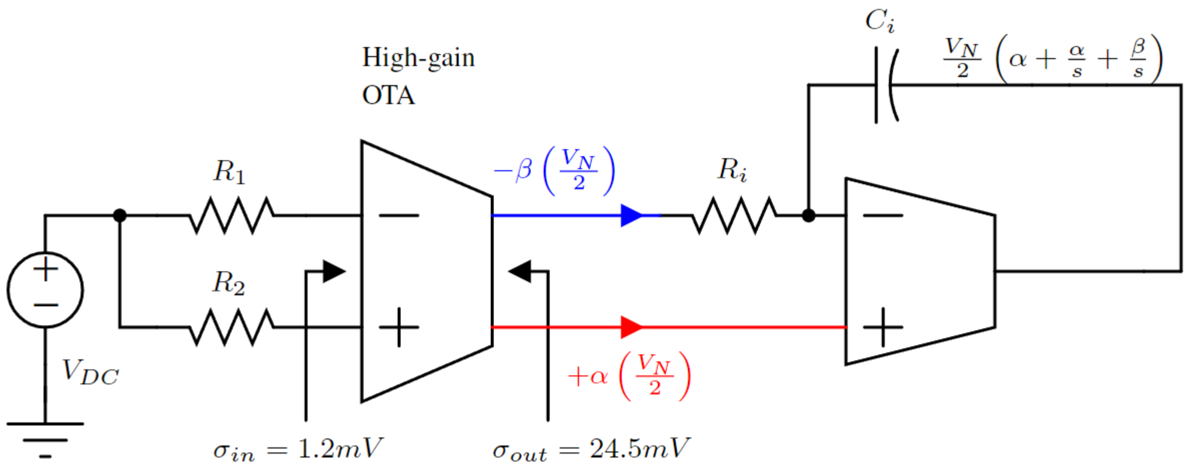
4-Bit CTDSM based TRNG

- 1st order, 4-bit CTDSM implemented using TSMC 65nm CMOS process.
- Thermal noise is generated by $R_{1,2}$, and its noise power is amplified using a High-gain OTA.
- Chaotic Ring oscillators (CRO) have been designed to generate clk_{adc} and clk_{dac} with amplified jitter.
- The two-phase operation of the Switched Capacitor RZ DAC ensures poor alias rejection.
- The LSB output from 4-Bit Flash ADC denoted as D_0 is sampled to generate raw random binary data, which subsequently undergoes for post-processing.



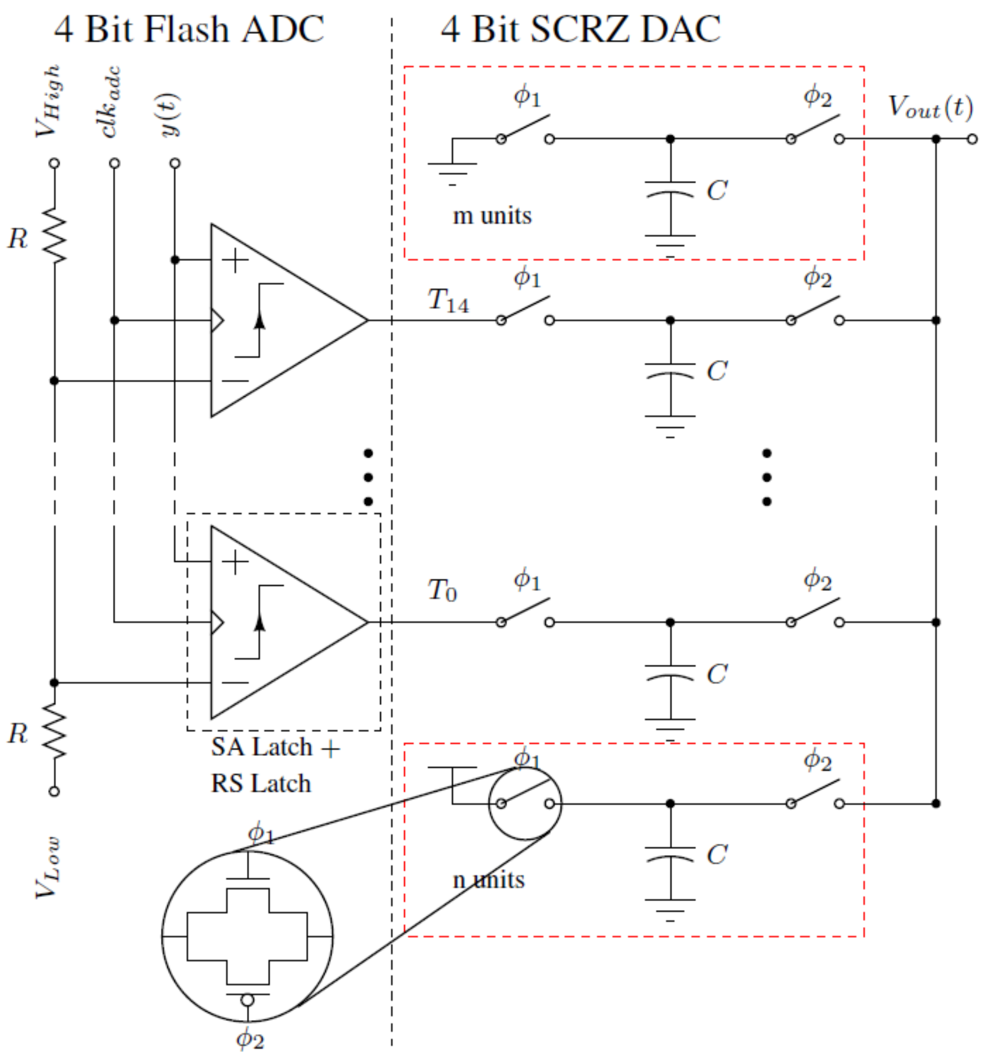
Thermal noise injection

- High-gain OTA, which calibrates a negative conductance load for impedance cancellation to achieve 54 dB gain and UGB of 5.4 GHz.
- OTA used in an open-loop configuration, where thermal noise from R_1 and R_2 served as inputs at a common mode voltage of V_{DC}.
- The differential output of the High-gain OTA becomes imbalanced because each terminal experiences a different load impedance.
- The reduction in the time constant of the integrator constrains the bandwidth of noise within the loop.



4-Bit ADC and DAC

- The design of a multi-bit quantizer to reduce the dynamic range to match the mean and standard deviation of the random process within the loop.
- Unit DAC element designed using a switched capacitor resistor with a Transmission gate as the switch.
- The dynamic range of both the ADC and DAC was calibrated to handle the measured full-scale swing of the noise input ($6\sigma_y$ = 440 mV) with an additional headroom of ±30 mV.
- The biasing for V_{High} and V_{Low} in the Flash ADC was set based on the value $V_{DC} \pm 3\sigma_y$.
- SCRZ DAC has always ON (m units) and always OFF (n units) DAC elements to adjust the dynamic range .
- The offset of the quantizer could be compensated by adjusting the number of always ON (OFF) DAC elements.



Chaotic Ring oscillator

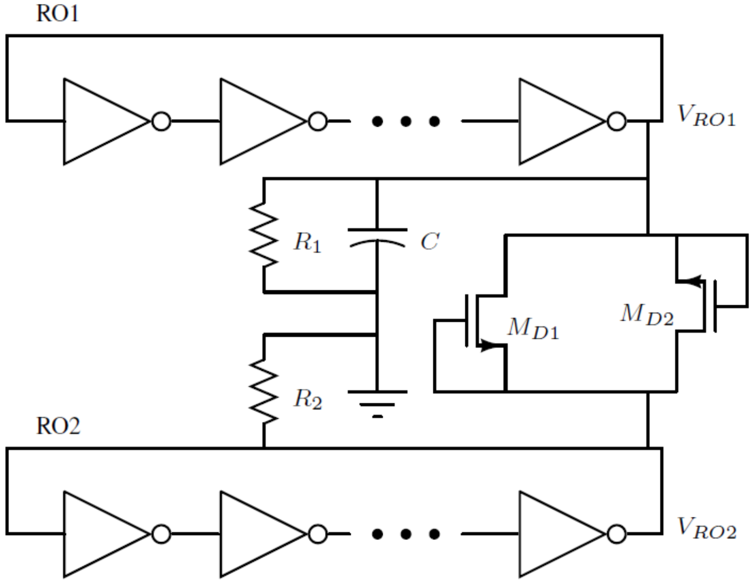


Figure: Diode Coupled Chaotic ring oscillator used to amplify the jitter.

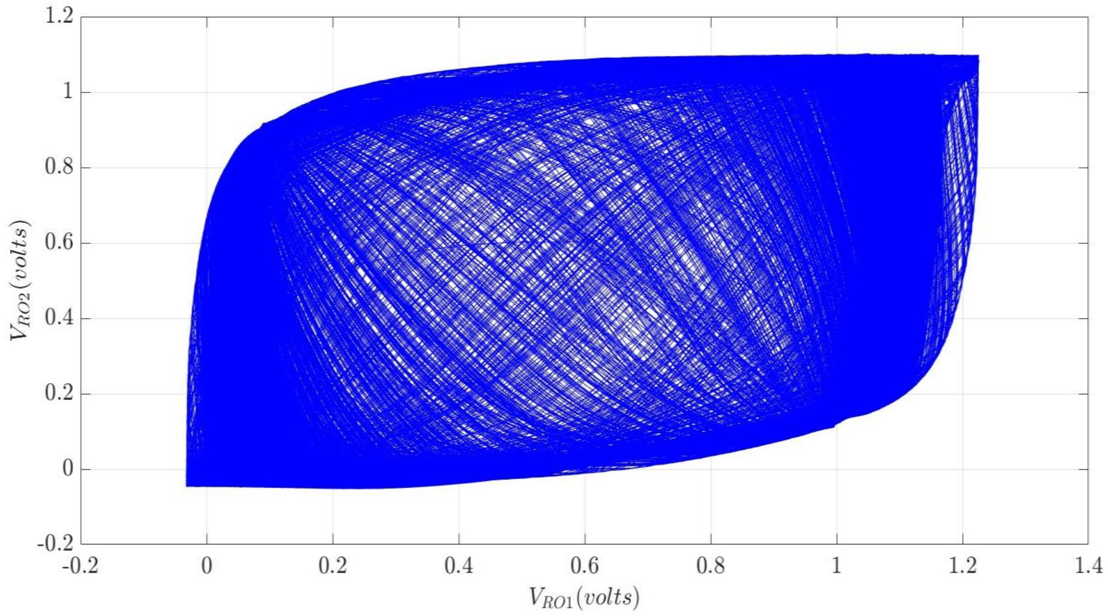


Figure: Phase portrait obtained by plotting V_{RO1} against V_{RO2} showing the chaotic behavior of the system, with no observable fixed pattern and unpredictable trajectories.

- Circuit boosts jitter using the chaotic dynamics generated by the non-linear coupling of two-ring oscillators.
- Measured clock frequencies: 98.4 MHz and 100.6 MHz, with a jitter-to-period ratio of approximately 20%.

Ihsan Cicek et al. “A chaos based integrated jitter booster circuit for true random number generators”. In: 2013 European Conference on Circuit Theory and Design (ECCTD). 2013, pp. 1–4.

Testing 4-bit CTDSM based TRNG

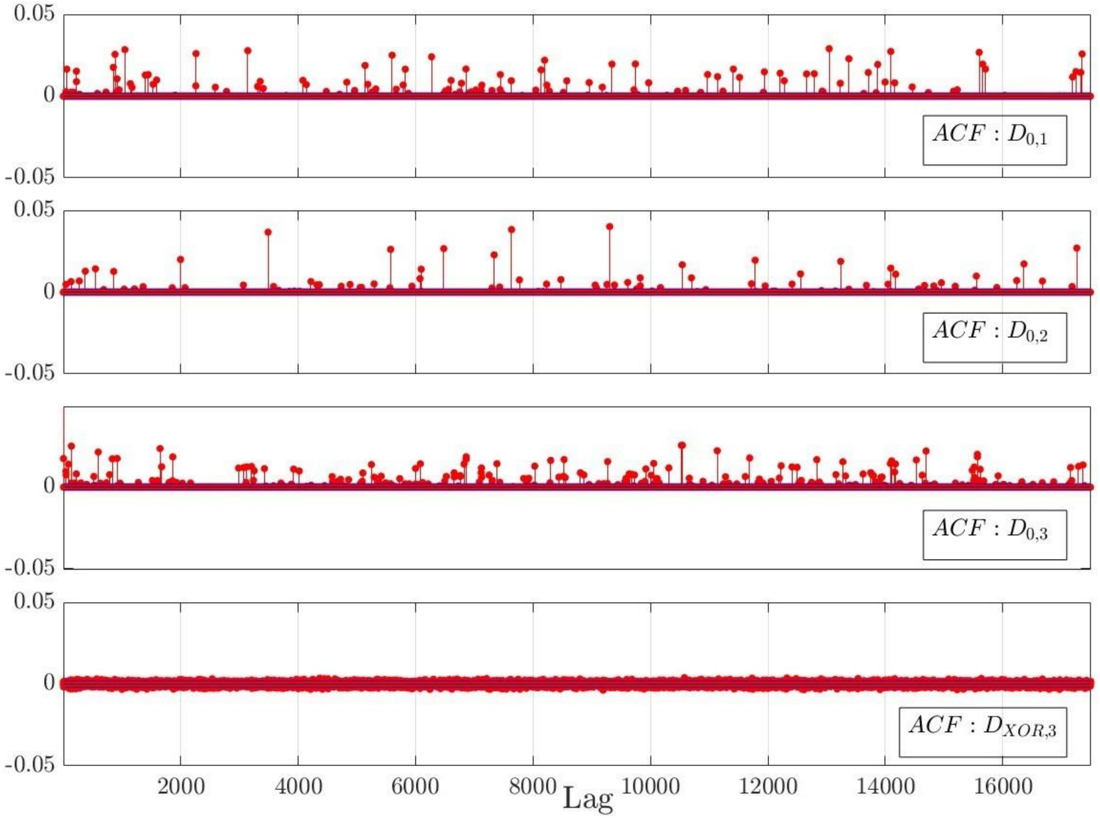


Figure: Autocorrelation function plotted for D0,x=1,2,3 and DXOR,3. We can observe the presence of micro-correlations in the raw, unprocessed data, whereas D0,x=1,2,3 shows no observable patterns.

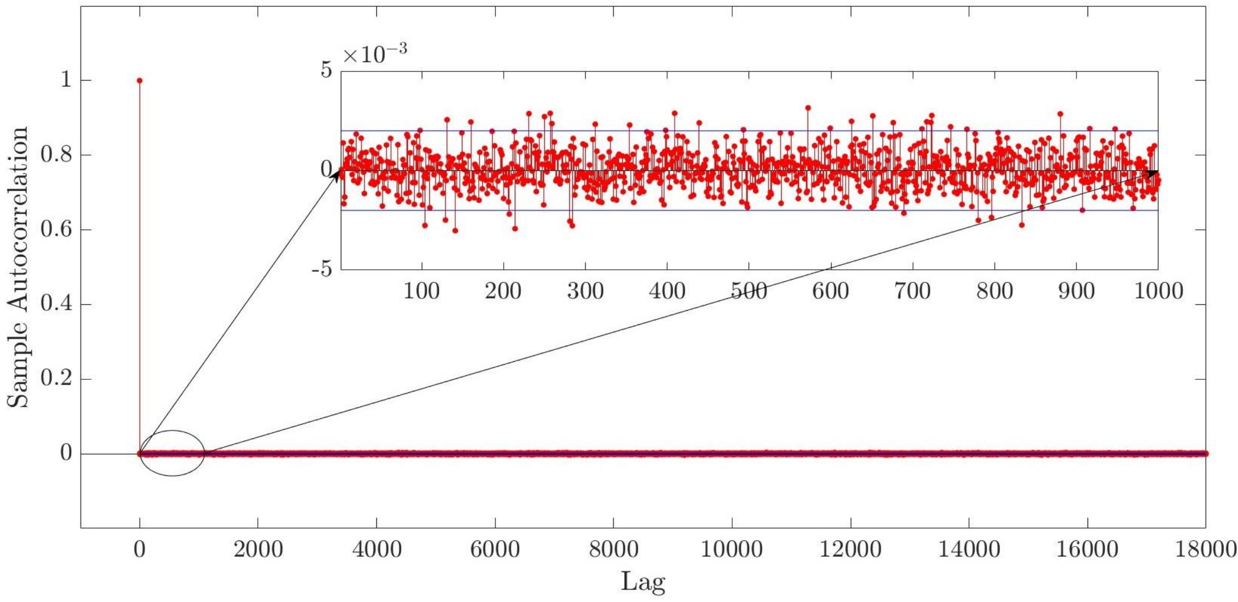


Figure: Autocorrelation plot for TRNG output data, DXOR,3 where ACF equal to 1 at Lag=0 and within ±0.002 with a 95% confidence bound for all non-zero Lag values.

Testing 4-Bit CTDSM based TRNG(cont.)

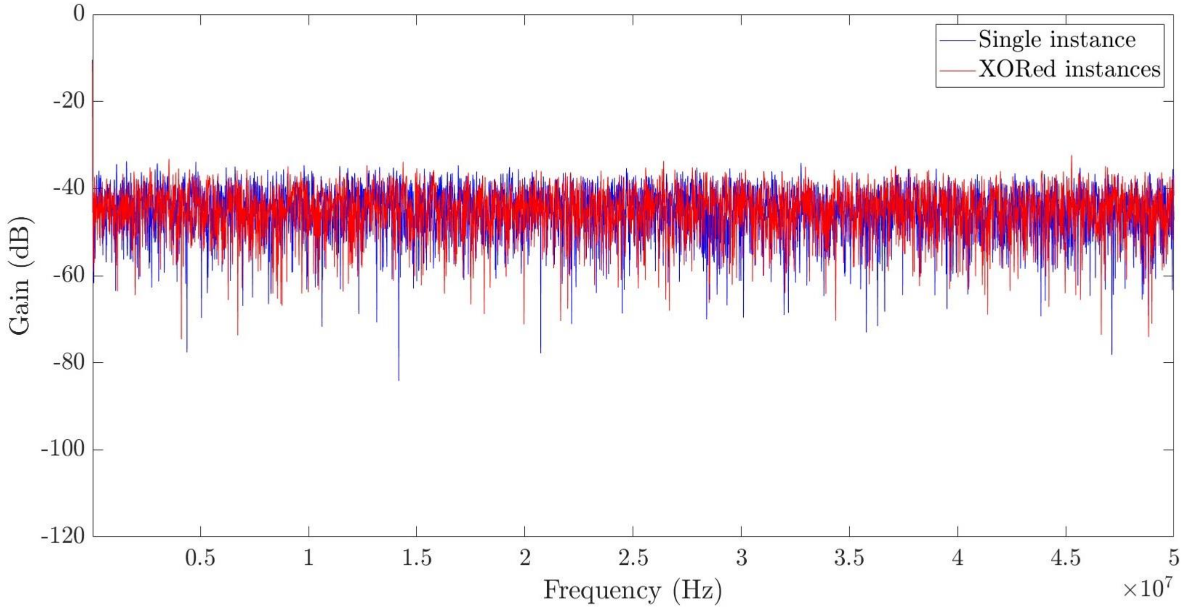


Figure: FFTs of D0,x and DXOR,3 both appears flat with no visible frequency dependence.

Serial No.	Test Name	Result
1.	Frequency monobit	Pass
2.	Block frequency	Pass
3.	Runs test	Pass
4.	Longest run of ones	Pass
5.	Binary matrix rank	Pass
6.	DFT	Pass
7.	Non-overlapping	Pass
8.	Overlap matching	Pass
9.	Maurers universal	Pass
10.	Linear complexity	Pass
11.	Serial test	Pass
12.	Approximate entropy	Pass
13.	Cumulative sums	Pass
14.	Random excursions	Pass
15.	Random excursions variant	Pass

Figure: One million binary random data samples of DXOR,3 passing all NIST 800-22 recommended test.

Conclusion

- Qualitative and quantitative analysis of thermal noise, quantization noise, clock jitter, metastability, and their impact on the noise floor of CTDSM.
- Theoretical justification for the design choices aimed at ensuring similar noise contribution from each noise source onto the overall noise signal variance within CTDSM.
- Circuit design techniques to boost thermal noise variance to inject it at the input of the CTDSM.
- An effective method of dithering white noise at the input of the ADC within the CTDSM loop.
- The design of a multi-bit quantizer to reduce the dynamic range to match the mean and standard deviation of the random process within the loop.
- Circuit design to generate a weighted combination of RZ and exponentially decaying pulses to leverage poor alias rejection and high jitter sensitivity simultaneously.
- The design of Chaotic Ring Oscillator to generate jittery clocks for ADC and DAC.

Future works

- **Device Variation Verification:** Thorough verification across Process, Voltage, and Temperature (PVT) induced device variations to ensure the robustness of the TRNG's random behavior.

- **Design Optimization:** Optimize the entire TRNG design to minimize power consumption while maintaining performance.

- **Environmental Impact Analysis:** Investigate the impact of environmental factors such as electromagnetic interferences, and power supply fluctuations on the entropy of the TRNG.

- **Post-Processing Enhancement:** More efficient post-processing techniques to enhance the quality of generated random numbers.